Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

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Abstract. **Keywords:** Dialect Normalization · Low-Resource NLP · Marathi Dialects · Sequence-to-Sequence Modeling · Synthetic Data Augmentation · mT5 · IndicBART · Text Standardization.

1 Introduction

Recent advances in deep learning and self-attention mechanisms [1] have driven rapid progress across natural language understanding and sequence-to-sequence generation tasks. The widespread adoption of Large Language Models (LLMs) [2] and open-weight foundational architectures [3] has enabled specialized text processing across domains such as finance, healthcare, and digital governance. However, current language technology ecosystems remain predominantly tailored toward standardized, high-resource national languages, creating a substantial performance bottleneck when applied to non-standard regional dialects [4]. Community surveys and global NLP evaluations [5,6] emphasize that native dialect speakers experience severe system degradation in downstream tasks—such as Automatic Speech Recognition (ASR), Machine Translation (MT), and conversational interfaces—due to unstandardized orthography, phonetic variation, and vocabulary shifts [7,8].

Dialect normalization—the task of transducing non-standard, regional dialectal text into normative written language while strictly preserving underlying semantic intent [8]—serves as an essential pre-processing layer to ensure digital linguistic inclusion. In multilingual environments such as India, which encompasses 22 scheduled languages alongside hundreds of unwritten regional dialects spoken by 1.4 billion people, sub-regional varieties remain drastically neglected. Marathi, an Indo-Aryan scheduled language spoken by over 83 million people in western India [9], provides a salient case study. While Standard Marathi (centered around the Pune literary register) possesses codified orthographic norms and digital infrastructure, regional sub-dialects—such as Southern Konkan (Malvani), Northern Konkan (Ahirani/Khandeshi), and Varhadi (Vidarbha)—exhibit marked phonological, lexical, and morpho-syntactic divergence from the normative written standard [10,11].

Building computational normalization systems for regional Indic dialects presents two core challenges: (1) *Extreme Data Scarcity*: parallel corpora mapping unstandardized regional dialects to normative standard forms are virtually non-existent; and (2) *Agglutinative Morphological Complexity*: Marathi features rich inflectional morphology, where noun case markers, postpositional affixes, and verbal tense/aspect suffixes undergo complex subword transformations rather than simple surface dictionary lookup [12, 13].

To address these limitations, this work establishes a multi-dialect Marathi text normalization benchmark utilizing synthetic corpus expansion and sequence-to-sequence transformer fine-tuning. Our core contributions are:

– **Curated Parallel Dialect Corpus**: Grounded in the RESPIN read-speech transcripts [14], we construct and release a parallel corpus mapping three major Marathi sub-dialects (Southern Konkan, Northern Konkan, and Varhadi) to Standard Marathi across agricultural and banking domains.
– **Automated Synthetic Augmentation with Multi-Tier Verification**: We design a domain-constrained synthetic generation engine using Gemma-4 [3] coupled with a multi-tier verification pipeline (Devanagari script integrity filtering and LLaMA-3.1-8B [2] semantic auditing) that purges corrupted pairs and expands the parallel corpus to 32,335 verified sentence pairs.
– **Cross-Validated Neural Benchmarks**: We conduct 5-fold cross-validated empirical evaluations comparing multilingual sequence-to-sequence architectures (`mT5-small` and IndicBART) across single-dialect, multi-dialect, original, and synthetically expanded dataset partitions, evaluating model scaling, verification engine ablation, and subword morphological error patterns.

2 Related Work

2.1 Text Transduction in Historical and User-Generated Corpora

Text normalization traditionally addresses two main forms of orthographic non-standardness: archaic spelling variation in historical manuscripts and unconstrained noise in digital User-Generated Content (UGC). Early historical normalization relied heavily on Levenshtein alignment, dictionary lookup, and Character-level Statistical Machine Translation (CSMT) [15, 16]. Subsequent evaluations across European languages demonstrated that subword tokenization via Byte-Pair Encoding (BPE) and sentence-level contextual modeling significantly reduce disambiguation errors compared to character-level transducers [17–19].

In the domain of digital communication (SMS, social media platforms, and online forums), orthographic noise exhibits complex combinations of informal slang, phonetic abbreviations, creative spellings, and unconstrained typos [20, 21]. Methodological paradigms transitioned from candidate edit-operation rankers like MoNoise [22] to standardized multilingual benchmarks such as MultiLexNorm [23], which confirmed that pre-processing via lexical normalization directly improves downstream dependency parsing and POS tagging. Recent work demonstrates that fine-tuning pre-trained byte-level models (byT5) and subword text-to-text transformers outperforms traditional statistical edit models for word-level sequence transduction [24, 25].

2.2 Regional Dialect Standardization across Language Families

Unlike UGC normalization, dialect-to-standard normalization maps coherent, systematic regional variations—encompassing distinct phonological, lexical, and morpho-syntactic features—into normative standard registers. In European languages, initial efforts benchmarked CSMT and statistical phrase-based models on Swiss German [26], which were subsequently surpassed by neural sequence-to-sequence architectures and pre-trained language models like `mT5-small` [27, 28]. Similar neural transduction approaches have been developed for Finnish, Swedish, and Estonian dialects [13, 29, 30]. A broad multi-lingual study across four language families by Kuparinen et al. [8] established that fine-tuning sequence-to-sequence transformers systematically outperforms rule-based and SMT baselines.

Beyond Europe, dialect processing has addressed structural divergence across East Asian and Afroasiatic varieties. Japanese regional dialects have been normalized using phrase-level LSTM transducers [31], while Mandarin studies focused on acoustic-phonetic mapping [32]. In Arabic NLP, the Conventional Orthography for Dialectal Arabic (CODA) framework [33] provided standardized guidelines for Modern Standard Arabic (MSA), paving the way for multi-city dialect translation benchmarks (MADAR [34], NADI [35]) and morphological rule-guided LLM prompting [36, 37].

2.3 Indic Dialect Landscape and Marathi NLP

Despite extreme linguistic heterogeneity and over 1.4 billion speakers, Indic regional dialects remain severely underserved in language technology [4, 6, 38]. While pre-trained multilingual sequence-to-sequence models such as IndicBART [9] and mT5 [25] cover major scheduled languages via unified Devanagari scripts or mC4 corpora, they lack domain-specific fine-tuning on regional sub-dialects. Monolingual initiatives like L3Cube-MahaNLP [39] offer Marathi encoder representations for classification, but do not support generative text standardization.

Recent benchmark initiatives such as INDIC-DIALECT [40] evaluate Hindi varieties (Bhojpuri, Magahi, Maithili), and WFST frameworks handle semiotic text normalization [41]. However, Marathi dialect processing has remained confined to acoustic/ASR classification [42, 43], read-speech transcripts (RESPIN [14, 44]), or zero-shot LLM retrieval [45, 46]. Our work fills this gap by establishing the first parallel corpus and 5-fold cross-validated sequence-to-sequence benchmark for regional Marathi dialects.

2.4 LLM Distillation and Multi-Tier Verification

To overcome severe parallel data scarcity, synthetic data generation using Large Language Models (LLMs) has gained widespread adoption [47–49]. Knowledge distillation strategies demonstrate that outputs generated by proprietary teacher models can effectively train compact student transformers [50–52]. However, unconstrained LLM generation for low-resource regional dialects introduces significant noise, including Hindi script leakage, entity hallucination, and unnormalized dialect residue. We extend synthetic augmentation by introducing a domain-constrained generation pipeline via Gemma-4 [3] coupled with a multi-tier verification engine (Devanagari script filtering and LLaMA-3.1-8B [2] semantic auditing), ensuring high data fidelity for transformer fine-tuning.

3 Methodology

3.1 Target Dialects and Linguistic Properties

We target the following major Marathi dialects that span the primary regions of Maharashtra and exhibit distinct linguistic and phonetic patterns:

1. **Southern Konkan (Konkan Coast)**: Characterized by shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms. Lexical differences include systematic vowel shifts: standard जातो ("he goes") becomes Southern Konkan जातां; standard कुठे ("where") becomes कुठंय. Domain-specific terminology generally remains untouched but often accompanied by Konkani words.
2. **Northern Konkan (Khandesh Region)**: Heavily influenced by neighboring Gujarati and Bhili language families,thus it exhibits significant consonant simplification and dental consonant shifts. Notabel transformations include: standard कशाला ("why") → Northern Konkan कसाले; standard मुलगा ("boy") → Northern Konkan पोर. Northern Konkan posseses the highest degree of lexical divergence among the three target dialects.
3. **Varhadi (Vidarbha/Eastern Maharashtra)**: Spoken in Vidarbha, It exhibits characteristic differences and verb suffix modifications. Standard interrogative का? ("why?") becomes Varhadi काऊन?; verbal forms undergo regular suffix mapping patterns. Among the three dialects, Varhadi maintains closest structural similarity to Standard Marathi as transformations are more systematic.

3.2 Original Corpus Curation and Foundation in RESPIN

Base sentences for our task are drawn from the domain-specific transcripts of the RESPIN-S1.0 initiative [14], developed at IISc Bangalore under the Gates Foundation project. While RESPIN has audio recordings and their transcripts across agriculture and banking domains for Indian languages and their dialects, it lacks aligned parallel dialect-to-target language translations required for efficient normalisation.

The large speaker diversity ensures broad coverage of intra-dialect phonetic and lexical variation, while the balanced gender and domain splits reduce systematic bias in our derived parallel corpus.

Table 1. RESPIN-S1.0 Marathi Source Corpus Statistics

Statistic	Value
Total Utterances	809,934
Total Duration	1,026.06 hours
Unique Speakers	2,644
Unique Pincodes	234

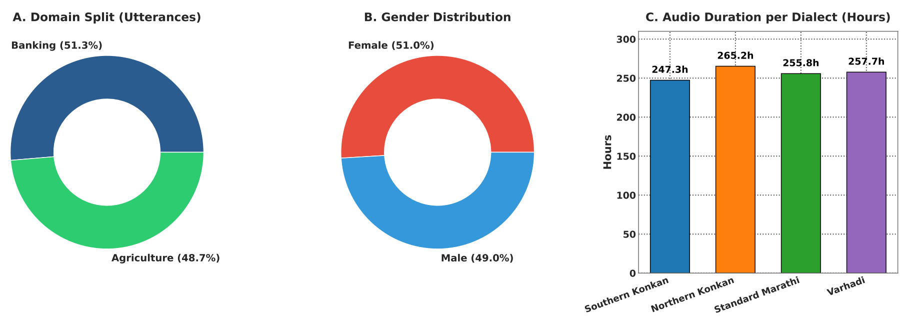

Fig. 1. RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution (49% Agriculture, 51% Banking); (B) gender split (51% Female, 49% Male); (C) audio recording duration per dialect variant.

3.3 LLM-Driven Parallel Corpus Generation and Verification

To deal with data scarcity, we developed an automated synthetic data generation pipeline powered by Gemma-4 [3] in 9B-instruction-tuned (`gemma-4`) configurations, accessed via Ollama. We considered several design decisions to ensure generation quality to be maximised:

- **1-Prompt-Per-Sentence Architecture**: Instead of generating batches of parallel pairs per API call (which risks array length mismatches and context contamination), each dialectal sentence is processed separately with a self-contained prompt.
- **Detailed Prompting Strategy**: The generation prompt enforces semantic fidelity rules: the standard output must preserve exact meaning, domain terminology (agriculture/banking), named entities, and numerical values from the input. The prompt includes dialect-specific lexical mapping tables and 5-8 few-shot examples per dialect.
- **Temperature Control**: Generation temperature is set to 0.3 with top-p = 0.9 to balance diversity with fidelity.

All generated pairs undergo a three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.** Automated validation checks implement Indic Abugida script normalization principles [12], including:

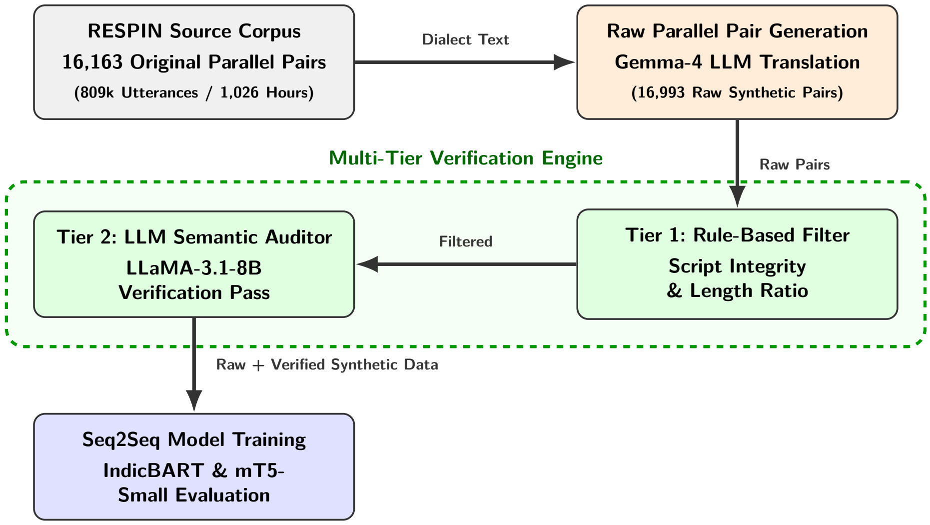

Fig. 2. System architecture of the proposed multi-dialect text normalization framework, illustrating LLM synthetic corpus expansion, multi-tier verification engine, and seq2seq model fine-tuning.

  - *Devanagari Script Integrity*: Both source and target must possess >90% Devanagari Unicode characters within the `U+0900--U+097F` range, automatically discarding unnecessary characters.
  - *Length Ratio Constraint*: $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
  - *Non-Identity Constraint*: $L_{\text{dialect}} \neq L_{\text{standard}}$ to exclude identical string copies.
  - *Minimum Token Count*: Both source and target must contain ≥ 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary LLM evaluator using `llama-3.1-8b-instant` (accessed via Groq API) performs semantic equivalence verification. The verifier rates on a 1-5 score scale considering: (a) semantic preservation; (b) absence of hallucinated entities; (c) absence of Hindi/English contamination; and (d) complete standardization of dialects. Pairs scoring < 4.0 are rejected or routed to correction.
3. **Closed-Loop Correction Engine.** Pairs flagged by Tier 1 or Tier 2 enter a two-step correction loop: Step 1 generates corrected translations using specialised prompts highlighting the specific failure reason (e.g., "Hindi leakage detected" or "Incomplete normalization"); Step 2 verifies the corrected sentences again,completely discarding pairs that fail a second audit pass.

As detailed in Table 2, out of 16,993 raw synthetic parallel candidate pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 corrupted or flawed pairs (an overall corruption/rejection rate of

Table 2. Parallel Dataset Composition, Synthetic Generation Yield, and Verification Filtering

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkan	5,576	5,838	5,569	269 (4.61%)	**11,145**
Northern Konkan	5,501	5,825	5,534	291 (5.00%)	**11,035**
Varhadi	5,086	5,330	5,069	261 (4.90%)	**10,155**
Total (All 3 Dialects)	**16,163**	**16,993**	**16,172**	**821 (4.83%)**	**32,335**

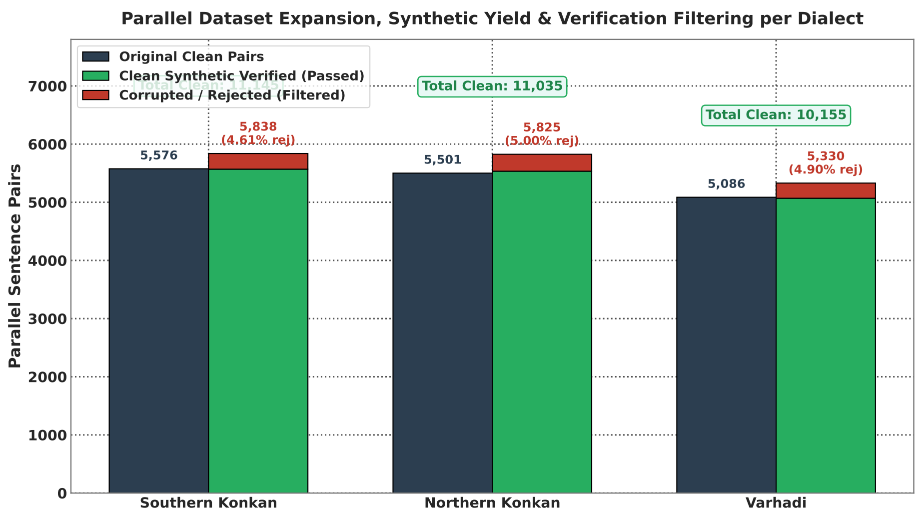

Fig. 3. Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

4.83%), yielding 16,172 clean verified synthetic pairs. Combined with the 16,163 clean original pairs, the final expanded benchmark suite comprises 32,335 high-quality parallel sentence pairs across all three dialects.

3.4 Evaluation Protocol

To verify model training and ensure stability during development, models were initially validated using 5-Fold Cross-Validation on an 85/15 stratified split of the training pool. For final benchmark reporting, models were trained on the complete dataset (combining original and synthetically expanded pairs) and evaluated on the official held-out RESPIN test set (`IISc_RESPIN_test_mr`, comprising 2,170 utterances across target dialects).

This setup is applied across all dataset configurations: Southern Konkan (Original), Northern Konkan (Original), Varhadi (Original), Combined-data (Original), and their Synthetically Expanded counterparts. All results are reported using standard automatic metrics computed via SacreBLEU [53] and Word Error Rate (WER) scripts:

Table 3. Comprehensive Evaluation Matrix on `IISc_RESPIN_test_mr` Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original (16k)	Southern Konkan	559	**57.80**	**26.60%**	43.86	34.37%
		Northern Konkan	540	**90.13**	**6.46%**	79.46	11.95%
		Varhadi	516	**83.59**	**10.19%**	74.81	14.73%
	Expanded (32k)	Southern Konkan	559	24.06	60.32%	**44.36**	**32.75%**
		Northern Konkan	540	65.41	28.70%	**79.76**	**12.61%**
		Varhadi	516	67.97	25.43%	**76.90**	**14.19%**
Multi-Dialect	Original (16k)	Southern Konkan	559	–	–	**40.94**	**35.34%**
		Northern Konkan	540	–	–	**77.50**	**14.65%**
		Standard Benchmark	555	–	–	**96.65**	**1.91%**
		Varhadi	516	–	–	**73.47**	**16.16%**
		Overall Combined	**2,170**	62.98	30.13%	**72.18**	**17.23%**
	Expanded (32k)	Southern Konkan	559	–	–	**43.88**	**34.96%**
		Northern Konkan	540	–	–	**78.16**	**13.55%**
		Standard Benchmark	555	–	–	**97.39**	**1.37%**
		Varhadi	516	–	–	**74.74**	**15.57%**
		Overall Combined	**2,170**	**76.50**	**16.23%**	73.48	16.58%

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, and smoothing method “exp”.
- **chrF++**: Character n-gram F-score combined with word unigrams and bigrams, capturing subword morphological variations in Marathi text.
- **Normalized WER / CER**: Word Error Rate and Character Error Rate computed after Devanagari Unicode script standardization.

4 Findings

4.1 Comprehensive Test Benchmark on `IISc_RESPIN_test_mr`

Key observations from Table 3 and Figure 4 include:

1. **mT5-Small Performance**: The synthetically expanded fine-tuned `mT5-small` model achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling**: Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, demonstrating that data expansion pulls out the latent capacity in mBART-50 architectures.

4.2 Single-Dialect vs. Multi-Dialect Benchmark Results on Original and Synthetic Datasets

Table 4 presents the performance matrix on 5-fold cross-validated original single-dialect benchmarks, and Table 5 explains the dialect-wise breakdown of the multi-dialect model on original data.

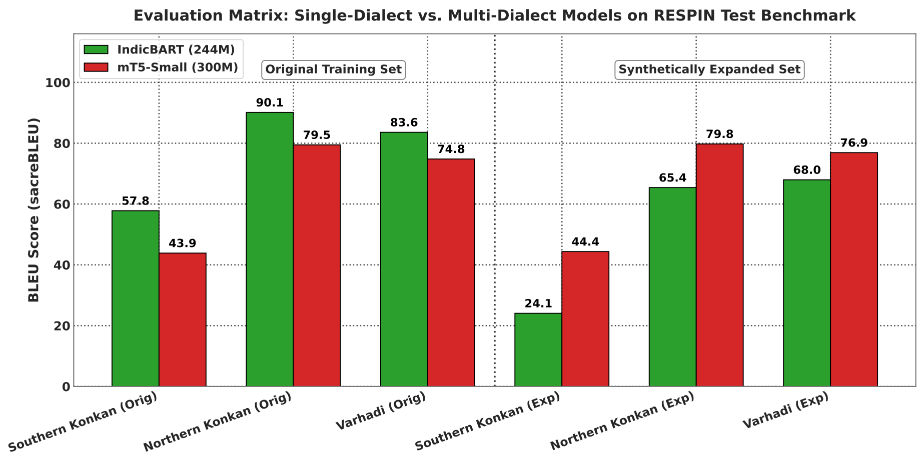

Fig. 4. Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

Table 4. Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models

Dialect Name	**Total Pairs**	**Test Set**	**IndicBART (244M)**		**mT5-Small (300M)**	
			BLEU	**chrF++**	**BLEU**	**chrF++**
Southern Konkan	5,576	836	**48.54**	**78.35**	46.31	74.43
Northern Konkan	5,501	825	**60.58**	**80.30**	60.31	77.34
Varhadi	5,086	762	76.63	90.93	**80.99**	**91.21**

Table 6 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 7 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data.

4.3 Impact of Synthetic Data Augmentation

Table 8 isolates the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit significant net gains (+8.95 BLEU for IndicBART, +6.22 BLEU for mT5-small), confirming that data scarcity was a primary performance bottleneck.

However, a fine-grained dialectwise breakdown reveals notable variance across sub-corpora. Southern Konkan benefits massively from expansion (+25.85 BLEU for IndicBART, +18.34 BLEU for mT5-small), while Northern Konkan sees consistent gains (+1.49 BLEU for IndicBART, +0.37 BLEU for mT5-small).

Table 5. Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown

Evaluated Subset	Total Pairs	Test Set	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan	16,163	837	26.21	53.15	**48.28**	**75.04**
Northern Konkan	16,163	826	47.59	67.16	**62.35**	**78.92**
Varhadi	16,163	763	72.75	85.63	**79.57**	**90.51**

Table 6. Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models

Dialect Name	Total Pairs	Test Set	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan	11,145	1,671	47.25	64.69	**65.10**	**82.88**
Northern Konkan	11,035	1,655	40.51	62.21	**62.07**	**79.29**
Varhadi	10,155	1,523	73.62	86.75	**78.89**	**90.59**

Varhadi displays slight baseline saturation variance (-2.48 BLEU for IndicBART, -0.84 BLEU for mT5-small) due to its already high original baseline performance. We hypothesize several tentative contributing factors for this differential impact:

1. **Baseline Saturation and Precision Dilution**: Varhadi already possessed a high original baseline (72.87–79.57 BLEU). We estimate that LLM-generated synthetic pairs may introduce minor stylistic variations or over-generalized suffix rules, slightly diluting exact N-gram matching against human reference texts.
2. **Lexical Drift and Subword Tokenization**: Northern Konkan exhibits distinct Gujarati and Ahirani lexical influences. We hypothesize that synthetic generation may introduce subtle lexical drift, which impacted IndicBART's narrower Devanagari subword vocabulary more noticeably than mT5-small's broader 250k multilingual SentencePiece vocabulary.
3. **Cross-Dialect Parameter Competition**: In multi-dialect training, we estimate that pooling synthetic data across distinct sub-dialects may cause capacity competition in shared attention weights, prioritizing recovery on severely sparse dialects (Southern Konkan) over already saturated partitions.

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

To quantify the explicit impact of our multi-tier LLM verification and filtering engine (`filter_flawed.py` and `llm_verifier.py`), we conduct a comparative

Table 7. Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialectwise Breakdown

Evaluated Subset	Total Pairs	Test Set	IndicBART (244M) BLEU	IndicBART (244M) chrF++	mT5-Small (300M) BLEU	mT5-Small (300M) chrF++
Southern Konkan	32,335	1,672	52.06	72.15	**66.62**	**83.57**
Northern Konkan	32,335	1,656	49.08	69.90	**62.72**	**79.63**
Varhadi	32,335	1,524	70.27	84.27	**78.73**	**90.46**

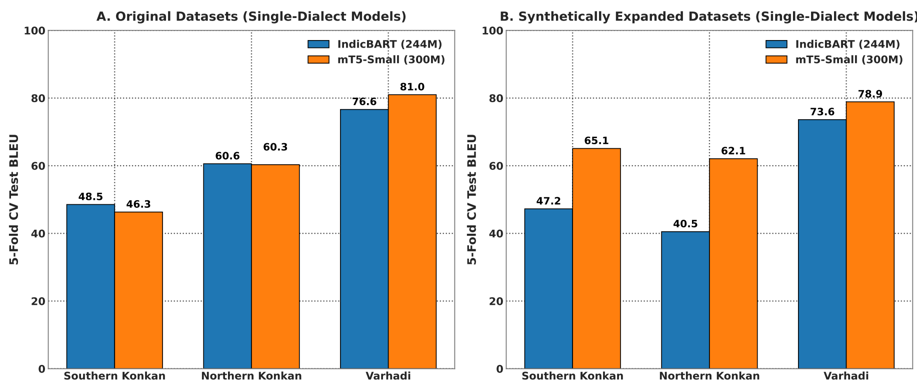

Fig. 5. Single-dialect 5-fold cross-validation test BLEU comparison for IndicBART (244M) and mT5-small (300M) on (A) Original datasets and (B) Synthetically Expanded datasets.

study evaluating models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Synthetic Data* (the expanded suite) across 5-fold cross-validation. The raw unverified dataset contains 19,914 parallel pairs, including 3,428 rejected or flawed pairs exhibiting Hindi language leakage, retained dialect residue, and garbled translations prior to multi-tier verification.

Table 9 presents the quantitative comparison on the benchmark suite.

Key findings and insights from this evaluation include:

1. **Quality Collapse on Unverified Data**: Training IndicBART on unverified raw LLM outputs causes a catastrophic 14.03 BLEU point drop (collapsing from 57.12 to 43.09) due to severe error propagation from uncorrected dialect residue, garbled syntax, and script errors.
2. **mT5-Small Resilience**: While google/mT5-small is more resilient due to its 250,000-token multilingual vocabulary, training on unverified data still suffers a heavy 8.30 BLEU point drop (falling from 69.51 to 61.21) and a 4.40 chrF++ drop.
3. **Essential Role of Multi-Tier Auditing**: Removing flawed pairs via `filter_flawed.py` and correcting recoverable generations via `correct_flawed.py` is directly re-

Table 8. Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	**Expanded**	**Δ**	**Original**	**Expanded**	**Δ**
Southern Konkan	26.21	52.06	**+25.85**	48.28	**66.62**	**+18.34**
Northern Konkan	47.59	49.08	**+1.49**	62.35	**62.72**	**+0.37**
Varhadi	**72.75**	70.27	−2.48	**79.57**	78.73	−0.84

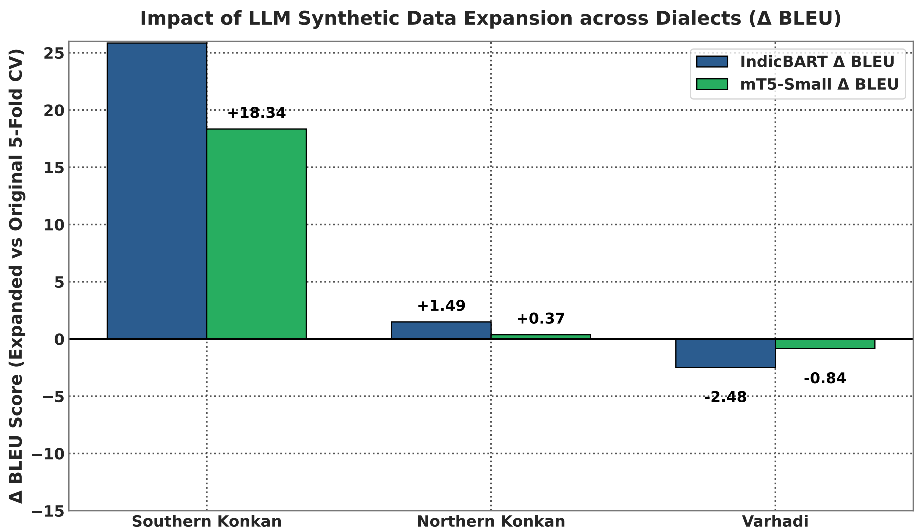

Fig. 6. Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

sponsible for eliminating hallucinated dialect markers, preserving semantic fidelity, and driving sequence-to-sequence normalization to state-of-the-art accuracy levels.

4.5 Model Architecture Analysis

Table 10 provides a detailed architectural comparison between the two primary models. The 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage**: mT5's 250K vocabulary provides 3.9× more subword units than IndicBART's 64K, enabling finer-grained tokenization of morphologically complex Marathi dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.

Table 9. Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	Pipeline State	Total Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	−14.03	−10.95
	Filtered Clean Data	32,335	**57.12**	**75.49**	**+14.03**	**+10.95**
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	−8.30	−4.40
	Filtered Clean Data	32,335	**69.51**	**84.58**	**+8.30**	**+4.40**

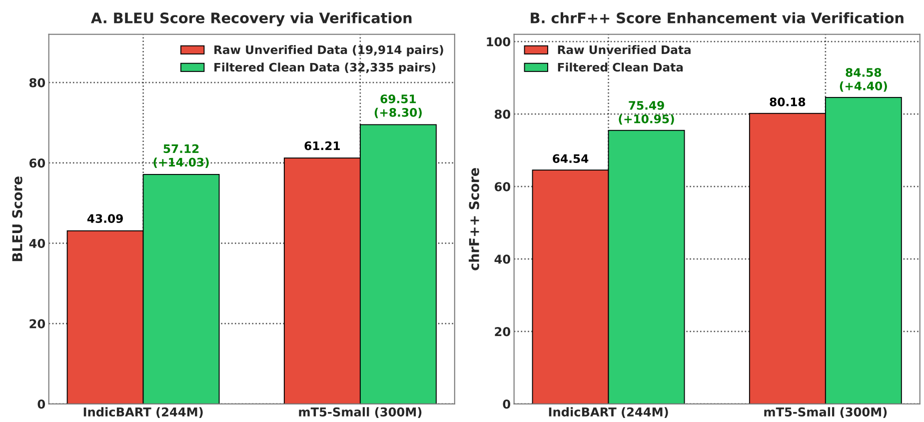

Fig. 7. Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

2. **Positional Encoding**: mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead**: IndicBART requires explicit language tokens (<2mr>) that add conditioning complexity, whereas mT5's pure text-to-text format enables direct mapping without task-specific architectural constraints.

5 Conclusion

This study presents a comprehensive, reproducible framework for multi-dialect Marathi text normalization, addressing severe data scarcity through automated synthetic data expansion and establishing rigorous neural sequence-to-sequence benchmarks.

Our LLM-driven synthetic data augmentation pipeline, powered by Gemma-2 with multi-tier verification (Devanagari script filtering and LLaMA-3.1-8B semantic auditing), successfully doubled the clean parallel corpus through synthetic expansion while maintaining strict semantic fidelity—a scalable paradigm

Table 10. Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	**IndicBART (244M)**	**mT5-small (300M)**
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required `<2mr>` token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

for other low-resource dialect pairs. Fine-tuning sequence-to-sequence transformers on the expanded benchmark established state-of-the-art performance: google/mT5-small (300M) achieved 69.51 BLEU and 84.58 chrF++ on the synthetically expanded multi-dialect dataset, outperforming IndicBART (244M) by +12.39 BLEU. Individual dialect performance reached 80.99 BLEU and 91.21 chrF++ for Varhadi, demonstrating that compact pre-trained sequence-to-sequence transformers can achieve high-accuracy dialect normalization.

5.1 Limitations

This work has several limitations that should be acknowledged:

1. Synthetic data generation relies on commercial LLM APIs (Gemma-2 via Google AI Studio), introducing API cost and rate limit considerations.
2. While we benchmarked two encoder-decoder architectures (mT5-small and IndicBART), decoder-only LLMs (e.g., LLaMA-3, Gemma-2) have not been evaluated for zero-shot or fine-tuned text normalization.
3. Evaluation is currently focused on text normalization metrics (BLEU, chrF++, WER); large-scale downstream task evaluation across diverse domain applications remains for future work.

5.2 Data and Code Availability

The complete parallel corpus of 32,335 verified dialect–Standard Marathi sentence pairs, along with all training code, evaluation scripts, and pre-trained model checkpoints, will be publicly released upon publication. The dataset will be available under an open license to support reproducible research in Indic dialect normalization. Our synthetic data generation pipeline, multi-tier verification engine, and fine-tuning configurations are documented in the accompanying code repository to enable replication and extension to additional Marathi sub-dialects and other low-resource Indic language varieties.

5.3 Future Directions

Several promising directions emerge from this work:

1. **Edge Deployment via Quantization**: Model quantization (INT8/INT4) and knowledge distillation to sub-100M parameter models for on-device deployment in rural Maharashtra.
2. **Expansion to Additional Dialects**: Extending the pipeline to Deshi/Puneri, Kolhapuri, Marathwada, and other Indic language sub-dialects.
3. **Decoder-Only LLM Benchmarking**: Evaluating instruction-tuned decoder-only models for zero-shot and few-shot dialect normalization tasks.

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